Literature Review Matrix: Gamification and Student Motivation

Field: Education | Years: 2021-2026 | Papers: 4 (supplied by the user) | Prepared 21 October 2026

#	Title	Author(s) & Year	Quartile	Method	Access	Introduction	Methodology	Results	Discussion	Theoretical Framework
1	Gamified quizzes and intrinsic motivation among Malaysian undergraduates	Tan & Rahman, 2023	Unranked/ Unavailable	Quantitative	Unclear	Tests whether gamified quizzes raise undergraduates' intrinsic motivation.	Cross-sectional survey; n = 320 undergraduates at two public universities; analysed with PLS-SEM.	Gamified quiz use significantly predicted intrinsic motivation (β = 0.38, p < .001).	Findings back SDT; authors advise designing quizzes that support learner autonomy.	Self-Determination Theory
2	Leaderboards in secondary science classrooms	Lee, 2022	Unranked/ Unavailable	Quantitative	Unclear	Tests the effect of leaderboards on engagement in secondary science lessons.	Quasi-experiment; n = 140 pupils in Penang; 8-week intervention.	Not retrieved (results section missing from the available copy).	Leaderboards may lower motivation among low-performing pupils.	Not explicitly stated
3	Students' lived experience of gamified learning	Ali & Wong, 2024	N/A (thesis)	Qualitative	Unclear	Explores how students experience courses that use gamification.	Interviews with 18 students; thematic analysis.	Three themes identified: competition, autonomy and fatigue.	Gamification fatigue emerges after week 6 of a course.	Flow theory
4	Badges and engagement in MOOCs: a review	Ali, 2021	Unranked/ Unavailable	Theoretical/ Review	Unclear	Reviews the evidence on digital badges and engagement in MOOCs.	Narrative review of 40 studies.	24 of 40 studies linked badges to short-term gains in course completion.	Whether badges have lasting effects remains unclear.	Goal-setting theory

Synthesis. Three sources each use a different theory (Self-Determination Theory, flow theory, goal-setting theory) and the leaderboard study names none, so there is no shared framework yet. Methods are spread out: two quantitative studies, one qualitative thesis and one narrative review, with no mixed-methods work. The evidence leans positive in the short term (a moderate SDT-consistent effect in the survey, short-term completion gains in 24 of 40 reviewed studies) but three sources point to limits: possible demotivation of low performers, fatigue after week 6, and unclear long-term effects. Gaps your study could fill: longitudinal designs past 6-8 weeks, mixed-methods work, and analysis of how prior achievement moderates the effect of competitive elements such as leaderboards. Two of the four studies are set in Malaysia, so findings from other settings are thin.

Notes on labels. *Quartile:* SCImago quartiles were searched for each journal on 21 October 2026. No SJR record was found for *Journal of Asian Education Studies*, *Science Teaching Review* or *Open Learning Letters*, and scimagojr.com could not be reached directly, so these are labelled Unranked/Unavailable instead of guessed. Paper 3 is a PhD thesis, so no quartile applies. *Access:* the DOIs could not be resolved and no open-access record was found, so access is Unclear for all four. *Paper 2:* its results section is missing from your copy, so that cell reads Not retrieved. All cells come only from the text copies you supplied, paraphrased.